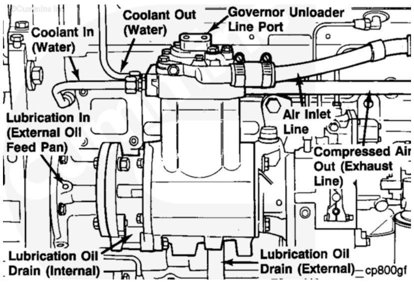
Trainings

General Information

Following the installation guidelines can greatly affect and enhance functional performance, operating efficiency, and useful product life of an engine accessory or brake system component.

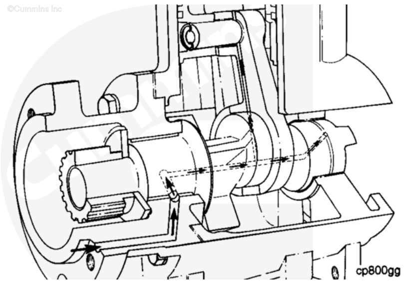
Following these recommended installation guidelines for Holset® air compressors will help provide reliable, service free operation well beyond the product warranty period.



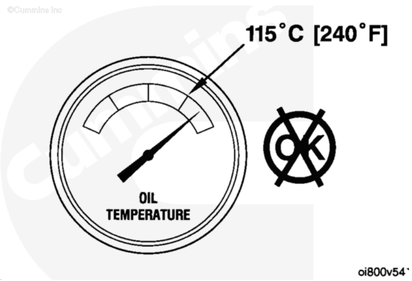
Lubricating Oil System

Holset® air compressors are lubricated with pressurized engine oil. The oil provides both a lubrication and cooling medium for bearings, piston rings, and cylinder walls.

For optimum air compressor performance, the following lubrication system guidelines are provided:



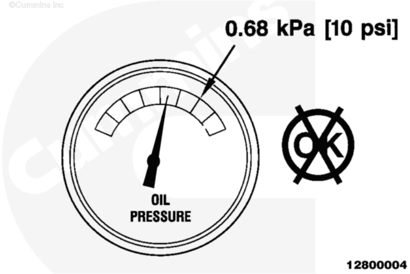
Oil temperature **must not** exceed 115°C [239°F] to provide proper lubrication capability. Higher operating temperatures will reduce cooling effects on the compressor and cause oil breakdown, and result in shortened air compressor life.



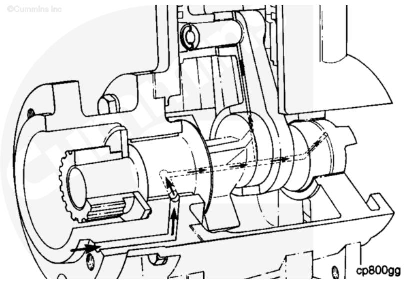
Maintain the following oil pressure:

kpa		psi
0.68	MIN	10

Lower oil pressure can result in a lack of lubrication on critical bearing surfaces, and shorten compressor life.



Engine oil supply is provided either internally or externally to the compressor. Flange mount models can have internal or external oil supply sources, while base mount models are provided with external oil supply **only**. When an external supply is used, a minimum of 3.18 mm [0.125 in] inside diameter (ID) line size **must** be used to provide adequate oil flow.

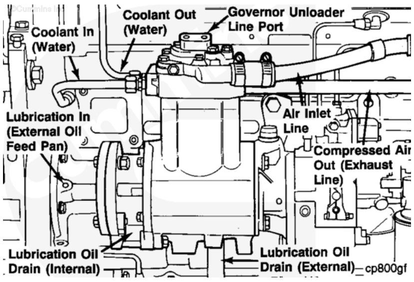


The supply line **must** be free of sharp bends or kinks which can cause flow restrictions. The oil supply source **must** be taken immediately after oil filtration from the engine for best results.

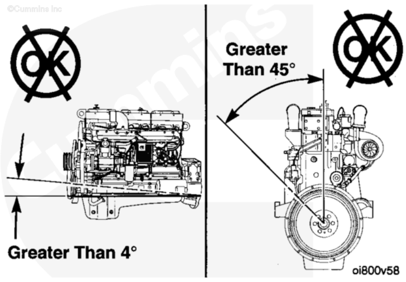
Use the oil drain interval specified by Cummins Inc. for a specific engine application. Cummins Inc. recommends the use of a high quality SAE 15W40 heavy-duty engine oil, such as Cummins® Premium Blue®, which meets the American Petroleum Institute (API) performance classifications CE or CF-4.

When operating in extreme environments where very high or very low temperatures or airborne dirt or dust is present, consider much shorter oil change intervals consistent with engine manufacturer recommendations.

Compressor oil drainage is critical to prevention of oil consumption. Flange mount compressor models provide either internal or external oil drainage. Base mount model provides external oil drainage **only**.

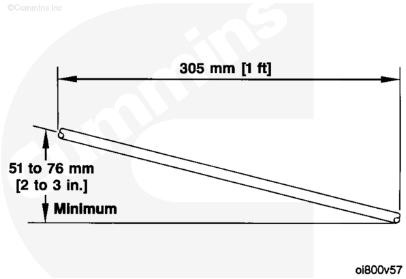


A 12.7 mm [0.50 in] inside diameter (ID) size is required when using an external oil drain. Using a smaller line size, or less than 12.7 mm [0.50 in] fittings, can cause excessive oil build-up in the compressor crankcase. The potentially damaging results include crankcase pressurization and oil passage into the vehicle's air system.

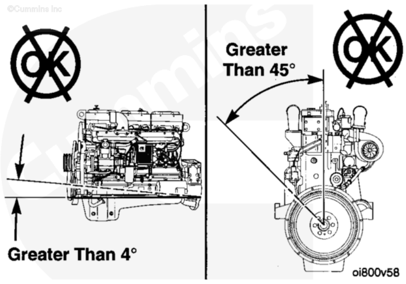


The oil drain system is gravity based, and therefore, requires installation of the drain line to be free of low spots or traps with a downward flow of 51 to 76 mm [2 to 3 in] per 305 mm [1 ft] minimum fall.

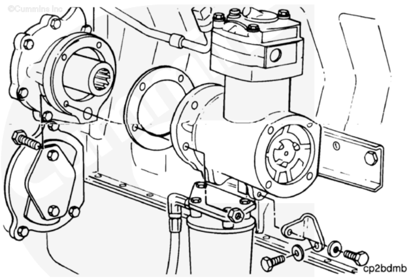
The drain line **must** enter the engine at the point above the engine oil level to prevent a back flow restriction.



Where flange mount compressor drainage is internal to the compressor with engine oil drain or return through engine block holes, vehicle installation tilt angles greater than 4 degrees from the horizontal or roll angles exceeding 45 degrees from the vertical position, can cause less than optimum drain back conditions.

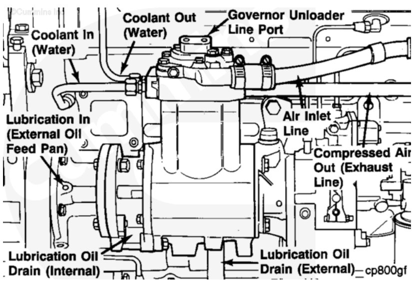


When mounting air compressors to the engine **always** use new OEM supplied gaskets to provide a leak free connection. Do **not** use form-in place or similar sealant materials as oil drain holes can become blocked.



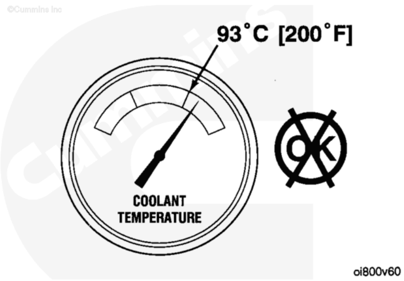
Cooling System

The engine cooling system provides the cooling water for Holset® air compressors. The air compressor functions produces high temperatures exhaust air, which if **not** controlled, causes excessive carbon formation in the compressor head and discharge or exhaust line. It can shorten compressor life.

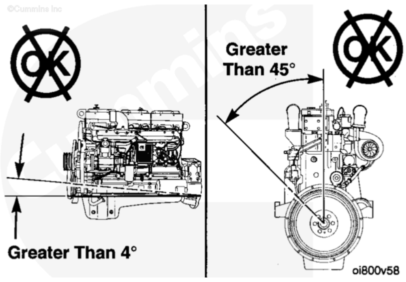


To provide adequate head cooling, a water flow rate of 16.9 liters [4.5 gal] plus or minus 3.785 liters [1.0 gal] per minute is required at rated engine speeds. Recommended minimum water flow rate through the head is 9.4625 liters [2.5 gal] per minute at rated engine speed with a 1:1 compressor drive ratio. Compressor to engine drive ratios higher than 1:1 can require higher coolant flow rates to provide optimum operating life. Contact a Cummins Authorized Repair location for further assistance under these conditions.

The engine's cooling water **must** be sourced immediately after the water pump to provide the lowest temperature water available. Water temperature to the compressor **must never** exceed 93°C [199°F].



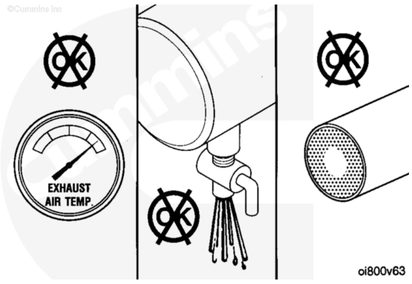
Inlet and outlet water line sizes **must** be 12.7 mm [0.50 in] outside diameter (OD) minimum of optimum flow rate. Either water port can be used for inlet or outlet water plumbing.



Normal air compressor life cycle is a function of application operating parameters and system installation configuration, both of which, along with maintenance practices, will influence compressor duty cycle.

Theory of Operation

Duty cycle is defined as the period of time the compressor is pumping (compressing air) as a percentage of total operating time. The air compressor duty cycle defines the life cycle, generally, lower duty cycles result in the longest life cycles.



Duty cycle observations range from 5 to 10 percent on new vehicles used in the line haul tractor applications to as high as 50 to 60 percent, or higher, for off-highway, garbage pick-up, or transit bus applications. Duty cycle can be measured on vehicles with either data recorders or by monitoring or timing the pumping cycle of the compressor (audible measurement) during vehicle operation.

Holset® air compressors are designed, for any given application, to have normal duty cycles of up to 25 percent and its life expectancy is defined with this duty cycle and the operating standard. The life of the air compressor will be progressively shortened as duty cycles continue to increase above 25 percent. If the duty cycle for specific application exceeds 25 percent, then additional compressor or system maintenance will be required or a larger size compressor **must** be specified to reduce the duty cycle.

High compressor exhaust air temperatures, excessive oil consumption, and carbon formation are typical characteristics of air compressors operating in excess of 25 percent duty cycle. If these symptoms are observed , corrective action **must** be taken immediately to reduce the duty cycle to a maximum of 25 percent. Failure to correct the cause of the high duty cycle (greater than 25 percent) can result in premature air compressor failure.

The following life cycle information is provided for all present Holset® product models:

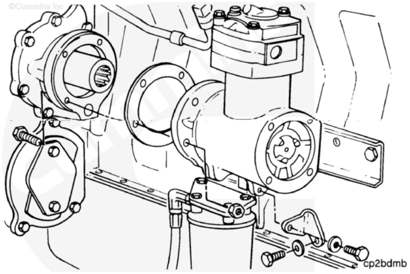
Duty Cycle	**Expected Life**
0 to 25 percent	Equal to or above 3 years, 300,000 miles, or 10,800 hours
26 to 40 percent	2 years or less
41 to 50 percent	1 year or less
Greater than 50 percent	Less than 1 year

The following application restrictions also apply:

- Maximum allowable system pressure set at 135 kPa [135 psi]
- Maximum turbocharged pumping cycle length = 1 minute (at maximum engine boost or rated engine speed)
- This information doe **not** alter or affect the written warranty for a specific unit, which is the **only** warranty that applies to a specific unit. Contact a Cummins Authorized Repair Location for written warranty details.

Installation Recommendations

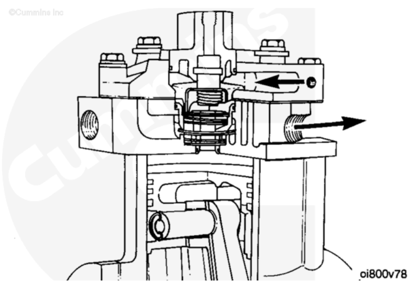
The following application information is generic and **must** be used **only** as a guide to the air compressor size required for a specific application. If vehicles are equipped with multiple air accessories, the next larger air compressor size is required.



Application	Air Compressor Model Recommended
Line Haul Tractor or Trailer	SS296 or QE296
Line Haul Doubles or Triples	SS338E or QE296
City Pick-up or Delivery	SS338E or QE296
Off-Highway or Construction	SS338E or QE296
Off-Highway or Mixer	QE338
School Buses or Rural	SS296 or QE296
School Buses or City	QE296 or QE338
Travel Coaches	SS338 or QE338
City Transit Buses	ST676
Residential Garbage Trucks	QE338 or ST676
Fire Trucks	QE338 or ST676
Bulk Haulers	QE338 or ST676
Recreational Vehicles (RV)	QE230 or QE296

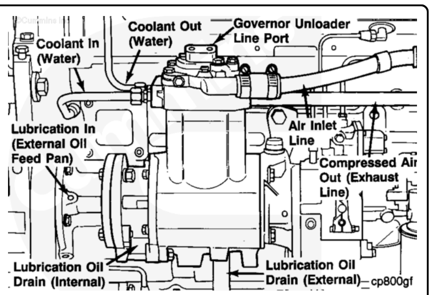
Air Intake System

For optimum compressor performance and system life, intake air **must** be clean and free flowing.



Two sources of intake air are available:

- Engine intake manifold (turbocharged engine applications)
- Air cleaner to the engine intake plumbing (naturally aspirated inlet air).



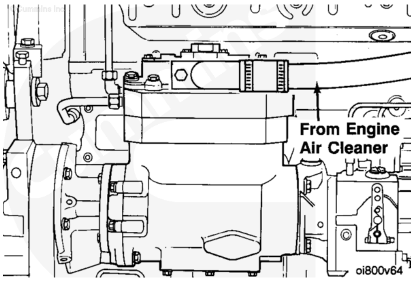
Optimum air pressure build-up time (shortest period) will result when using the engine intake manifold as an air supply source. Utilizing this source of air, the compressor intake will be pressurized by the engine turbocharger system.

Holset® recommends turbocharged inlet air, sourced from the engine manifold, for all air compressors operating under normal duty cycle applications.

All Holset® air compressors are designed for the full engine operating range of turbocharged (pressurized) inlet air without use of pressure reducing or restrictive devices.

All engines can utilize a naturally aspirated (ambient non-pressurized) intake air source. For best performance, this air **must** be sourced from the engine air cleaner directly.

Holset® recommends naturally aspirated inlet air sourced directly from the engine air cleaner for twin cylinder high-duty cycle applications.



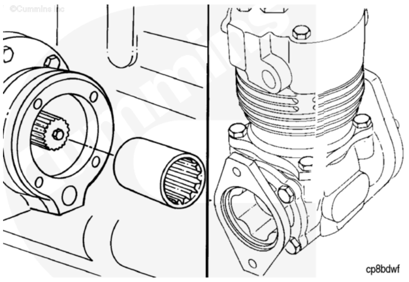
Drive Units

Holset® air compressor can be driven by either of two options:

- Gear driven via axial load, spline coupling combination, or
- Side load direct drive, meshed gears.

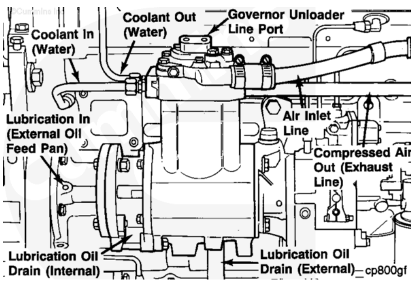
All flanged mounted compressors are gear driven.

Compressor through-drive options are available for power steering and fuel pumps on some engine applications.

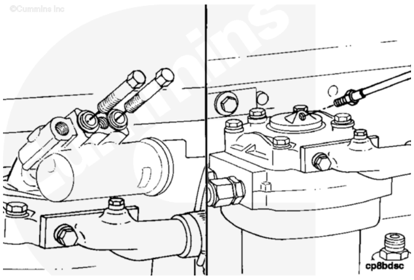


Compressed Air System

Air governors, air dryers, and air tanks are key functional components in the air system. The following recommendations will minimize air system problems and maximize the life of the air compressor and the air system.



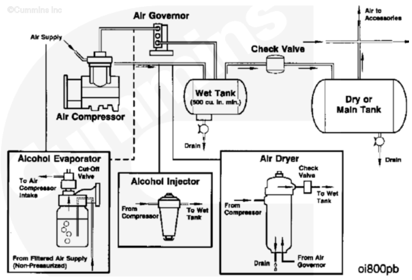
The air governor provides the signal to actuate and terminate compressor pumping cycles.



The governor mounts on the compressor governor mounting pad. If **not**, and remote mounting is required, locate the governor as close as possible to the compressor. Using a signal line between the governor and compressor exceeding 1.06 m [3.5 ft] is **not** recommended and can result in unloader actuation problems.

Avoid mounting the governor in high heat areas of the engine. Rubber seals in the governor can deteriorate and cause failures under high temperature exposure. Environmental temperatures exceeding 93°C [199°F] **must** be avoided.

Air dryers are recommended for all vehicle air brake systems. The primary function of the air dryer is moisture removal to prevent downstream freeze-ups and corrosion of air lines, air tanks, and valving components.



The air dryer also functions as oil and air contamination removal system, which helps provide improved system performance and longer service life.

Consult the manufacturer for optimum air dryer installation requirements.

Holset® does **not** recommend alcohol injectors and evaporators. Typically, these devices use a wide variety of alcohol mixtures which can cause negative reactions with downstream components, and shorten system life.

Use a an evaporator system with turbocharged compression inlet air can require an additional valve to prevent siphoning the alcohol into the compressor air intake and subsequently into the engine.

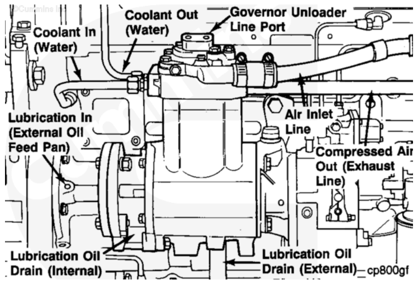
Use of an injector system adds inherent restrictions in the compressor discharge line, potentially increasing build-up times and carbon formation problems.

Air tanks provide the vehicle with an air storage reservoir area for braking needs. As a minimum, a service and emergency air reservoir tank are required. A third tank is also provided and acts as a reservoir wet tank where moisture and other system contaminates condense during the air cooling process downstream of he compressor.

To obtain optimum air system performance, all air reservoir tanks **must** be purged daily to prevent excess accumulation of contaminate material and reduced air storage volume.

The air discharge line provides the route for compressor high-pressure air from the air compressor to the air storage or reservoir tank system.

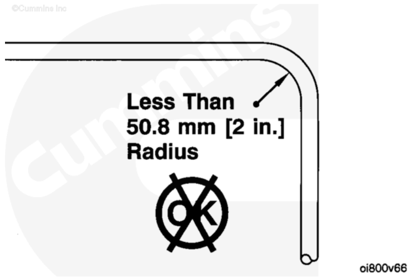
As a result of compressor discharge air containing a small amount of oil mist at high temperature, carbon formation in the discharge line becomes a common air system problem.



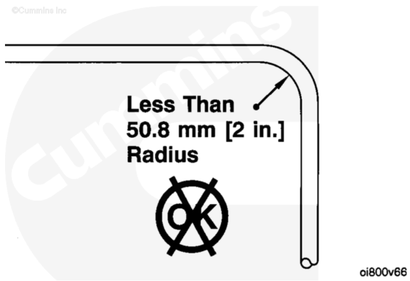
In severe instances, the problem can lead to discharge line restrictions and shorten compressor life. To prevent this problem, the following installation guidelines **must** be followed:

- To provide air cooling of discharge air, the plumbing line connected to the exhaust port **must** be made of copper, steel, or wire braided Teflon tubing capable of withstanding 1034 kPa [150 psi] pressure peaks and normal continuous line pressures 931 kPa [135 psi] at temperatures of 232°C [450°F]. Ideally, 1.8 m [6 ft] minimum length of copper tubing, or 3.1 m [10 ft] minimum of Teflon or stainless steel tubing provides adequate air cooling.

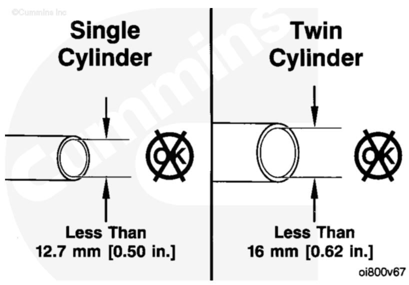
If an air dryer is used, see manufacturer's guidelines.



- To minimize air flow restriction caused by carbon formation, avoid discharge line bends with a radius of less than 50.8 mm [2 in].
- General air flow direction **must** be downward from the discharge port without traps where oil and moisture can collect. Discharge line low spots or traps can result in carbon formation or water freeze points which restrict air flow.

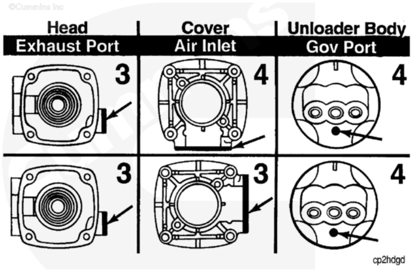


- Discharge line sizes for the single cylinder compressors **must** be 16 mm [5/8 in] inside diameter (ID) or a number 12 hose, with a minimum of 12.7 mm [1/2 in] inside diameter (ID) or a number 10 hose for optimum performance.



- Holset® twin cylinder compressor optimum line size is 19 mm [1 in] inside diameter (ID) or number 16 hose with a minimum of 16 mm [5/8 in] inside diameter (ID) or number 12 hose.
- Smaller than optimum line sizes will reduce compressor life under high duty cycle conditions.

The Holset® single cylinder compressor's unique design allows multiple independent porting options for plumbing flexibility in the field. Both the top cover (containing the inlet air port) and the head (containing the water and air discharge ports) can be independently rotated 360 degrees, in 90 degree increments.



Additionally, the unloader body mounted on top of the compressor can be rotated 360 degrees in 90 degree increments, independent or the top cover and head, to allow governor mounting orientations four times.

Note : When orientation is changed in the field, new gaskets must be used when the air compressor is assembled.

Last Modified: 13-Jun-2005